

Similarity and complex vector spaces

Definition

For A and B , two $n \times n$ matrices, we say that A is similar to B if there is an invertible matrix P such that $P^{-1}AP = B$. ✓

Definition

- $\underline{\mathbb{C}}^n$ is a *complex* vector space where addition is coordinatewise and so is scalar multiplication but now with complex numbers.
- For a complex $n \times n$ matrix A , we say that $\underline{\lambda} \in \mathbb{C}$ is an eigenvalue of A if for some non-zero $\underline{u} \in \underline{\mathbb{C}}^n$, $\underline{A}\underline{u} = \underline{\lambda}\underline{u}$.

Uses of similarity

$$\underline{P^{-1}AP = B} \quad \text{for some invert. } P \text{ and matrices } A \text{ and } B.$$

Compute B^{100}

$$\underbrace{B \cdot B \cdot B \cdot B \cdots B}_{100 \text{ times.}} = \underbrace{P^{-1}A \cancel{P} P^{-1}A \cancel{P} P^{-1}A \cancel{P} \cdots}_{100 \text{ times.}} P^{-1}AP$$

$$= P^{-1} \underline{A^{100}} P$$

For instance if A is diagonal, $A = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

$$A^{100} = \left(\begin{array}{c} \lambda_1^{100} \\ \vdots \\ \lambda_n^{100} \end{array} \right)$$

$$\underline{B^{100}} = \underline{P^{-1}} \underline{A^{100}} \underline{P}$$

Ex: $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \leftarrow$ has char. poly. $\lambda^2 + 1$

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ -1 & -\lambda \end{pmatrix} \\ = \lambda^2 + 1.$$

eigenvalues are $\pm i$

and the eigenvectors.

$\lambda = i$: Solve $\begin{pmatrix} -i & 1 \\ -1 & -i \end{pmatrix} u = 0$, $u = t \begin{pmatrix} 1 \\ i \end{pmatrix}$.
 $t \in \mathbb{C}$. $i u_1 = u_2$

$\lambda = -i$: Solve $\begin{pmatrix} i & 1 \\ -1 & i \end{pmatrix} u = 0$, $u = t \begin{pmatrix} i \\ 1 \end{pmatrix}$.
 $u_1 = i u_2$.

$$\text{Let } P = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad \det(P) = 1 - i^2 = \underline{\underline{2 \neq 0}}$$

$$\underline{AP} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}$$

$$P \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}$$

$\uparrow$
 D

$$AP = \underline{P} D \text{ so } \underline{P^{-1}AP = D.}$$

$$\begin{aligned} A^2 &= P^{-1} D^2 P, & D &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \\ &= -P^{-1} I P & D^2 &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I. \\ &= -I \end{aligned}$$

Diagonalization

Theorem

An $n \times n$ matrix A is similar to a diagonal matrix iff there are n linearly independent eigenvectors for A . In fact, if $P^{-1}AP = D$ where D is a diagonal matrix then the columns of P represent n linearly independent eigenvectors.

Diagonalizable

If A is similar to a diagonal matrix then we say A is diagonalizable.

Imagine we have $v_1, \dots, v_n$ lin. indep eigenvectors
for A corr. to $\lambda_1, \dots, \lambda_n$.

$$P = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

$$\underline{AP} = \begin{pmatrix} | & & | \\ \frac{Av_1}{1} & \frac{Av_2}{1} & \dots & \frac{Av_n}{1} \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ \lambda_1 v_1 & \dots & \lambda_n v_n \\ | & & | \end{pmatrix}$$

$$PD = \begin{pmatrix} 1 & & \\ v_1 & \dots & v_n \\ 1 & & 1 \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$$\text{where } D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$$= \begin{pmatrix} 1 & & \\ \lambda_1 v_1 & \lambda_2 v_2 & \dots & \lambda_n v_n \\ 1 & & & 1 \end{pmatrix}$$

$$= AP$$

P is invert. because $v_1, \dots, v_n$ were lin. indep.

$$\text{so } D = P^{-1}AP.$$

The general case

Theorem

- *If A is a complex $n \times n$ and A has n eigenvalues (counting multiplicities) then A is diagonalizable iff for every eigenvalue, the algebraic multiplicity equals the geometric multiplicity i.e. for every eigenvalue λ , the algebraic multiplicity of λ equals the dimension of the λ -eigenspace.*
- *Every symmetric real matrix has only real eigenvalues and is diagonalizable.*

Corollary (Very important special case)

If an $n \times n$ matrix A has n different eigenvalues then A is diagonalizable.

For: $A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$ $\det(A - \lambda I) =$

$$(1-\lambda)(2-\lambda)(3-\lambda)$$

Eigenvalues are $\lambda = 1, 2, 3$.

$\lambda = 1$: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \underline{e_1}$ is an eigenvector for $\lambda = 1$.

$\lambda = 2$: Solve $(A - 2I)u = 0$ so $u = z \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ for $z \in \mathbb{C}$

$\lambda = 3$: Solve $(A - 3I)u = 0$ so $u = z \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ for $z \in \mathbb{C}$.

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\text{So } P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & 0 & 2 \end{pmatrix} \quad \text{then } \cancel{P^{-1}}AP = \underline{\underline{\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}}}$$

Step-by-step diagonalization algorithm

- 1 Suppose $\lambda_1, \dots, \lambda_k$ are the eigenvalues of A and their algebraic multiplicities are $n_1, \dots, n_k$.
- 2 For each $i = 1, \dots, k$, determine a basis B_i for the λ_i -eigenspace of A . If for any i , the dimension of this eigenspace is less than n_i then A is not diagonalizable.
- 3 Otherwise, collect together all the bases, $B_1, \dots, B_k$ and place these vectors in the columns of a matrix P .
- 4 P will be invertible and $AP = PD$ where D is diagonal with diagonal entries $\lambda_1, \dots, \lambda_k$ repeated $n_1, \dots, n_k$ many times respectively.